



Assignment 01

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Subject	SCD

1. Requirement Specification Document:

➤ Functional Requirements

1. User Management

1.1 Add New User (Librarian/Student/Member)

- **Purpose:** Register new users (librarians, students, or members) into the system.
- **Details:**
 - Collect details like **name, ID, contact info, and role**.
 - Assign a unique **User ID** for identification.
 - Store user data in a database.

1.2 Edit/Update User Information

- **Purpose:** Modify existing user details (e.g., phone number, address).
- **Details:**
 - Search for a user by **ID or name**.
 - Allow updates to personal information.

1.3 Delete User

- **Purpose:** Remove a user from the system (e.g., if they leave the institution).
- **Details:**
 - Search for the user and confirm deletion.
 - Ensure no active book loans before deletion.

1.4 View User Details

- **Purpose:** Display user information for administrative purposes.
- **Details:**
 - Search by **ID, name, or role**.
 - Show borrowing history if applicable.

2. Book Management

2.1 Add New Book

- **Purpose:** Enter a new book into the library database.
- **Details:**
 - Input **title, author, ISBN, genre, publication year, and quantity**.
 - Assign a unique **Book ID**.

2.2 Update Book Information

- **Purpose:** Modify book details (e.g., change quantity, correct author name).
- **Details:**
 - Search by **title, ISBN, or Book ID**.
 - Allow editing of book metadata.

2.3 Remove Book

- **Purpose:** Delete a book from the system (e.g., if lost or damaged).
- **Details:**
 - Ensure no active loans before removal.
 - Log deletion for records.

2.4 Search Books

- **Purpose:** Allow users to find books easily.
- **Details:**
 - Search by **title, author, ISBN, or genre**.
 - Display availability status (**Available/Borrowed**).

3. Borrowing & Returning System

3.1 Issue Book (Check-out)

- **Purpose:** Allow users to borrow books.
- **Details:**
 - Verify user eligibility (no overdue books).
 - Record **check-out date** and **due date**.
 - Update book status to "**Borrowed**".

3.2 Return Book

- **Purpose:** Process book returns.
- **Details:**
 - Scan **Book ID** or enter manually.
 - Check for **late returns** and apply fines if needed.
 - Update book status to "**Available**".

3.3 Renew Book

- **Purpose:** Extend borrowing period if no reservations exist.
- **Details:**
 - Check if renewal is allowed (max renewals = 1 or 2).
 - Update **due date**.

3.4 Track Overdue Books

- **Purpose:** Identify late returns and manage penalties.
- **Details:**
 - Automatically flag overdue books.
 - Calculate fines (**e.g., \$1 per day late**).
 - Send reminders via email (if implemented).

4. Reporting & Analytics

4.1 Generate Borrowing Reports

- **Purpose:** Track book circulation trends.
- **Details:**
 - Show **most borrowed books**.
 - Filter by **date range, genre, or user type**.

4.2 View Fines & Payments

- **Purpose:** Manage late fees and payments.
- **Details:**
 - List users with pending fines.
 - Record payments and update balances.

4.3 Inventory Status

- **Purpose:** Check total books, available copies, and lost/damaged books.
- **Details:**
 - Display summary statistics.
 - Export data to **CSV/PDF** (optional).

5. Authentication & Security

5.1 Login/Logout

- **Purpose:** Secure access to the system.
- **Details:**
 - **Librarians** have full access.
 - **Members** can only view/borrow books.

5.2 Password Reset

- **Purpose:** Allow users to recover accounts.
- **Details:**
 - Email verification or security questions.

➤ **Non-Functional Requirements**

1. Performance:

- System should handle **100+ concurrent users**.
- Search results should load in **<2 seconds**.

2. Security:

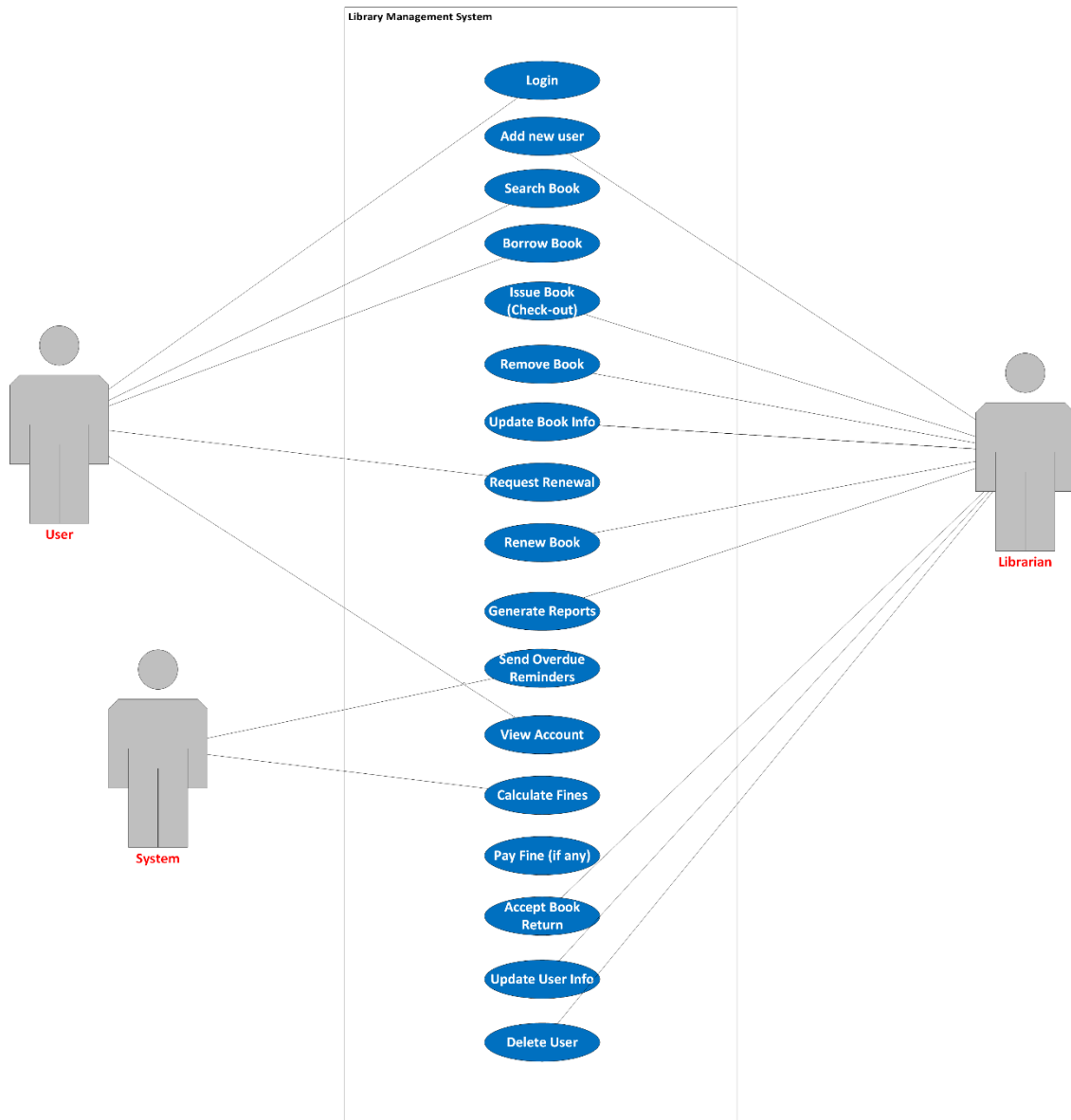
- Passwords must be **encrypted**.
- Unauthorized users **cannot modify data**.

3. Usability:

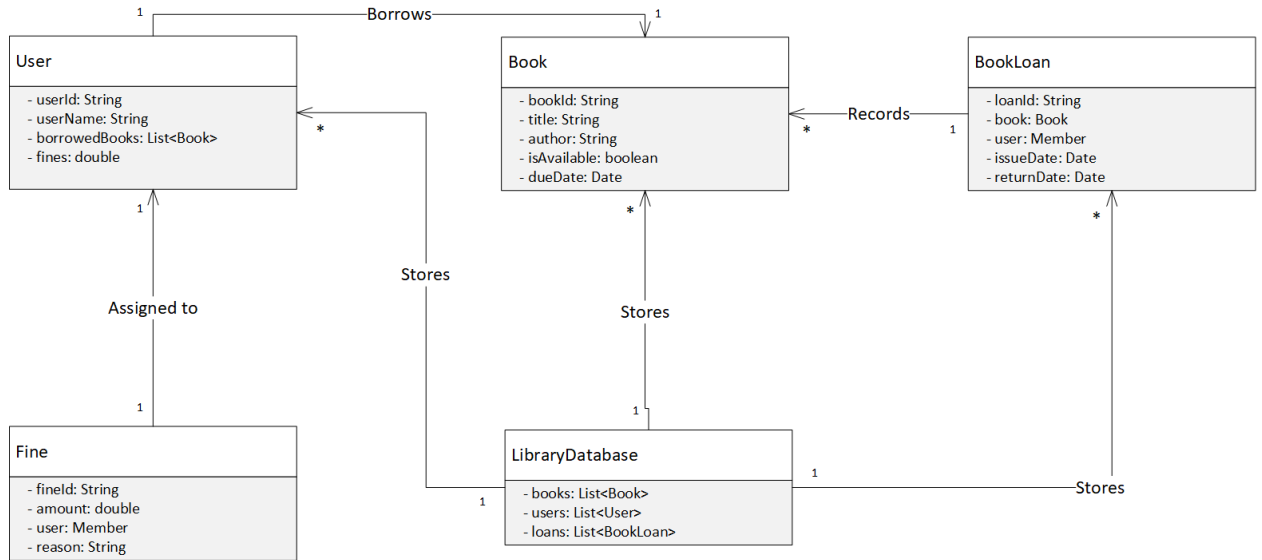
- Simple, intuitive interface.
- Mobile-friendly (optional).

2. Detailed Design Document:

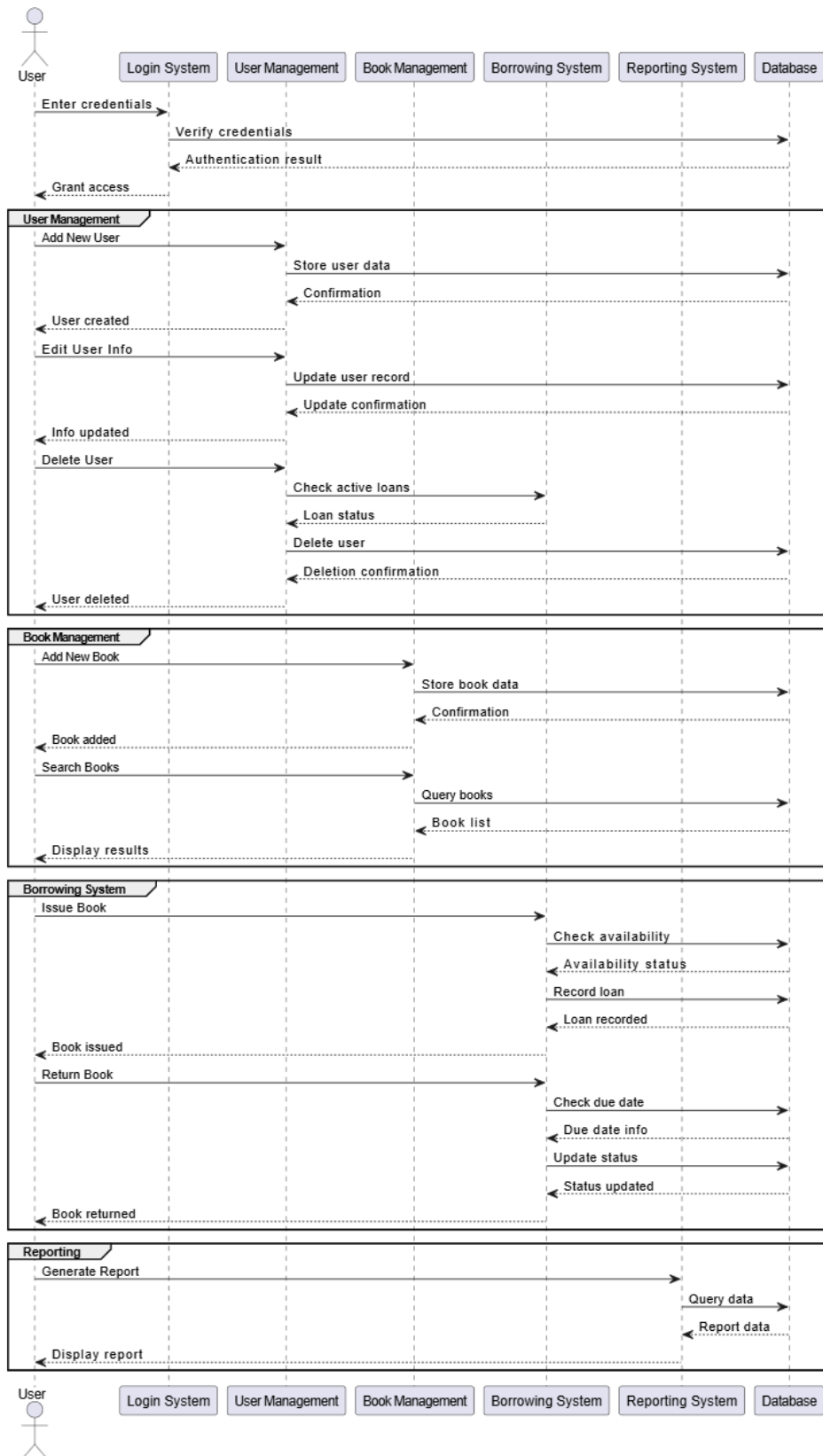
- Use-case diagram:



- Object diagram:



• Sequence diagram:



- **Communication diagram:**

